STUDENT PROJECT REPORT

GridGuard

ESP32 voltage, current and power protector with a relay cut-off

Submitted by

Team members _____

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Department / Class _____

College / Institution _____

Project guide _____

Academic year _____

> **Mains voltage**
> This project switches 230 V AC. Work on it only when it is unplugged, keep every module inside a closed box, keep the MCB as the real protection, and never connect USB while it is on mains.

1. Introduction

GridGuard sits between the wall socket and an appliance. It measures the mains voltage and the load current continuously and keeps a relay on the live wire closed only while voltage, current and power are all inside limits that the user chooses. If one of them stays outside its limit for 2 seconds, the relay opens and the reason is logged. Everything is shown on a colour TFT and on a web page served over GridGuard's own WiFi hotspot, where the limits can be set, the relay switched and the readings calibrated.

Aim: to protect appliances such as fridges, pumps and chargers from bad supply voltage and from overload or dry-running, with clear local and phone feedback, using low-cost parts.

2. Specifications

Table 1. GridGuard specifications

Item	Value
Supply and load	230 V AC, 50 Hz, single phase; load up to 6 A
Switching	5 V relay module, 10 A contacts, live wire, normally open (no power = load off)
Protection limits	lower (LSL) and upper (USL) limit for voltage, current and power, any values with LSL below USL; LSL 0 = not used
Default limits	voltage 190–250 V, current 0–6 A, power 0–1300 W
Trip delay	2 s (a value must stay outside its window this long)
After a voltage trip	automatic: relay closes 10 s after the voltage is back inside its window
After a current or power trip	locked off until reset (button hold 5 s or web ON)
Start grace	current and power LSL checked from 5 s after the relay closes
Measurement	1600 samples per channel in 200 ms (8 kHz); V, I, P, PF five times a second
Display	1.77" ST7735S TFT, 128 x 160, SPI; five screens; Bai Jamjuree font
Controls	push button (screens, reset, relay off); web ON/OFF switch
Indicator	red LED in the relay input: LED ON = relay closed (socket powered), LED OFF = relay open
Web	own WiFi "GridGuard" (password gridguard), http://192.168.4.1: live values, 60 s graph, limits table, relay switch, calibration, trip log
Kept after power-off	limits and calibration (ESP32 flash)

3. How it works

Figure 1 (next page) shows the blocks. The live wire runs from the plug through the MCB, the relay contacts, the current sensor and the rocker switch to the socket. The voltage sensor and the 5 V supply are connected after the MCB but **before** the relay, so GridGuard keeps running and keeps measuring the supply after it has cut the load off.

3.1 Sensing and maths

- **Voltage:** ZMPT101B transformer module; **current:** ACS712-20A Hall sensor in the live wire (100 mV/A).
- Both outputs pass a 10 kΩ / 20 kΩ divider (5 V to 3.3 V) with a 100 nF filter into GPIO34 and GPIO35 (ADC1, works with WiFi on).

- Every 200 ms the ESP32 reads 1600 samples per channel (10 mains cycles), removes the DC midpoint and computes Vrms, Irms, real power P = mean(v × i) and power factor. Windows stretched by WiFi (outside 196–204 ms) are dropped.

3.2 Limits and states

Each quantity has a window from LSL to USL. Voltage is checked all the time. Current and power can only be seen while the relay is on, so they are checked only then, and their LSL (under-current, under-power, for example a pump running dry) only from 5 s after the relay closes, to let a motor start. Figure 4 in Section 5 shows the state machine.

Table 2. Protection states

State	Relay	Meaning and what happens next
WAITING	off	power-up or after a voltage trip; counts 10 s while the voltage is inside its window, then closes the relay
ARMED	on	everything inside its window; watching
TRIPPING	on	a value is outside its window; counting 2 s. Back inside in time: ARMED again
TRIPPED	off	relay opened, trip logged; shown for 3 s
LOCKED	off	after a current or power trip; stays off until reset (button hold 5 s or web ON)
OFF (manual)	off	turned off by hand (button hold 2 s or web OFF); stays off until turned on

3.3 Setting the limits and resetting

- **Set the limits** on the web page, section Limits: type LSL and USL for voltage, current and power (0 as LSL switches a lower limit off) and press **Save limits**. They are saved in flash.
- **After a voltage trip** nothing needs to be done: the relay closes by itself 10 s after the voltage is back inside its window.
- **After a current or power trip** the relay stays off (screen: HOLD 5s RESET). After fixing the cause, **hold the button 5 s** (the screen counts RESETTING 5 4 3 2 1; letting go cancels) or tap **ON** on the web page. The relay closes and every limit is checked again.
- **Relay off by hand:** hold the button 2 s while the relay is on, or tap OFF on the web page.

3.4 Relay drive and indicator

Most 5 V relay modules are **active-low**: IN at 0 V switches the relay on, IN left unconnected keeps it off. To switch them off through a wire, IN must be pulled almost to 5 V, which a 3.3 V ESP32 cannot do. GridGuard therefore pulls GPIO26 to 0 V for ON and releases the pin (nothing driven) for OFF, and a **red LED in series with IN** takes up about 1.8 V so that a released pin really means OFF. The same LED is the indicator: **red LED ON = relay closed (socket powered), red LED OFF = relay open.** Because the pin is released while the ESP32 boots, resets or is being flashed, the relay is always off then.

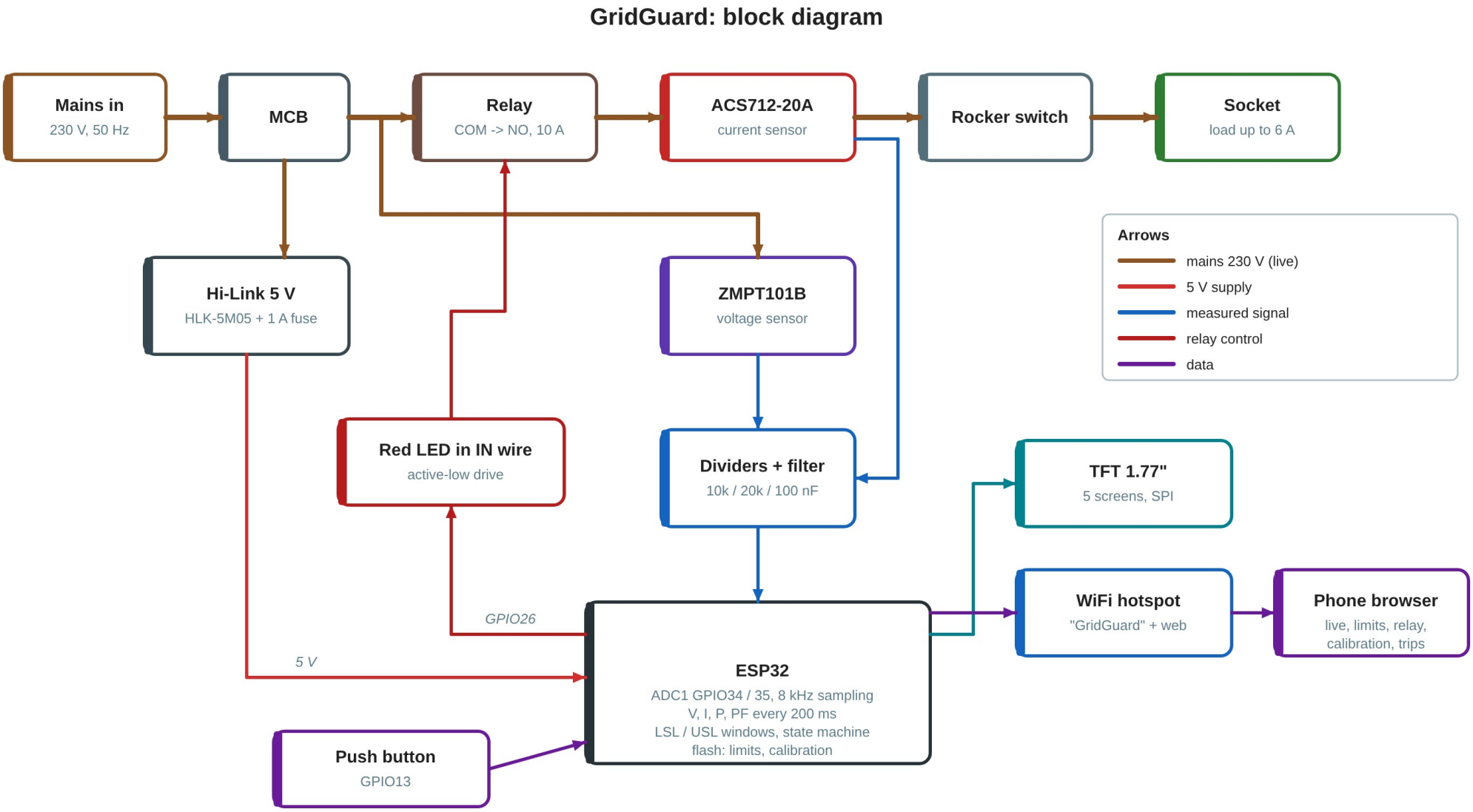


Figure 1. *GridGuard block diagram*

4. Components list

Table 3. Components

#	Component	Specification	Purpose
1	ESP32 DevKit	38-pin, ESP-WROOM-32, CP2102 USB	controller, WiFi, web server
2	Voltage sensor	ZMPT101B module, 5 V	measures mains voltage
3	Current sensor	ACS712-20A module, 5 V (100 mV/A)	measures load current
4	TFT display	1.77" ST7735S, 128 x 160, SPI (8 pins incl. BLK)	local display
5	Relay module	5 V, 1 channel, 10 A, relay SRD-05VDC-SL-C	switches the live wire
6	Red LED	5 mm red, in series with relay IN	makes the relay switch off; relay-on indicator
7	Push button	12 mm tactile or panel push button	screens, reset, relay off
8	Power supply	Hi-Link HLK-5M05, 5 V 1 A	powers the electronics
9	Fuse + holder	1 A slow-blow	protects the Hi-Link input
10	Resistors	10 kΩ 1 % x2, 20 kΩ 1 % x2	5 V to 3.3 V dividers
11	Capacitors	100 nF ceramic x2	divider noise filter
12	MCB	single pole, modular	short-circuit protection
13	Socket, rocker switch	6 A modular	load socket, manual ON/OFF
14	Enclosure	modular plate + surface box	housing
15	Plug and cord	6 A 3-pin plug, 3-core cord	mains input

5. Wiring

5.1 Pin map

Table 4. ESP32 connections

ESP32 pin	Goes to	Note
5V / VIN	5 V rail (Hi-Link +Vo)	no series diode
GND	GND rail	common ground
3V3	TFT VCC and BLK	3.3 V output
GPIO34	voltage divider node	ADC1, input only
GPIO35	current divider node	ADC1, input only
GPIO23	TFT SCK	SPI clock
GPIO19	TFT SDA	SPI data (MOSI), not I2C
GPIO5	TFT CS	
GPIO4	TFT RST	
GPIO27	TFT DC	
GPIO26	relay IN through a red LED	LED long leg to IN, short leg to GPIO26; no resistor
GPIO13	push button to GND	internal pull-up, no resistor

Table 5. Module connections

Module	Pin	Connect to
ZMPT101B	VCC / GND / OUT	5 V / GND / 10 kΩ to the GPIO34 node; L / N input after the MCB, before the relay
ACS712-20A	VCC / GND / OUT	5 V / GND / 10 kΩ to the GPIO35 node; IP+ / IP− in the live wire after the relay
Relay module	VCC / GND / IN	5 V / GND / red LED to GPIO26; COM to live from the MCB, NO to ACS712 IP+
TFT	VCC, BLK / GND / CS, RST, DC, SDA, SCK	3V3 / GND / GPIO5, 4, 27, 19, 23
Hi-Link	AC L / AC N; +Vo / −Vo	live after the MCB via 1 A fuse / neutral; 5 V rail / GND

> **Note**
> Dividers: 10 kΩ from the sensor output to the node, 20 kΩ and 100 nF from the node to GND. Each node reads about 1.65 V with nothing flowing.

5.2 Diagrams

Figure 2 shows every low-voltage connection on the real 38-pin pin order. Figure 3 shows the 230 V path. Figure 4 shows the protection states.

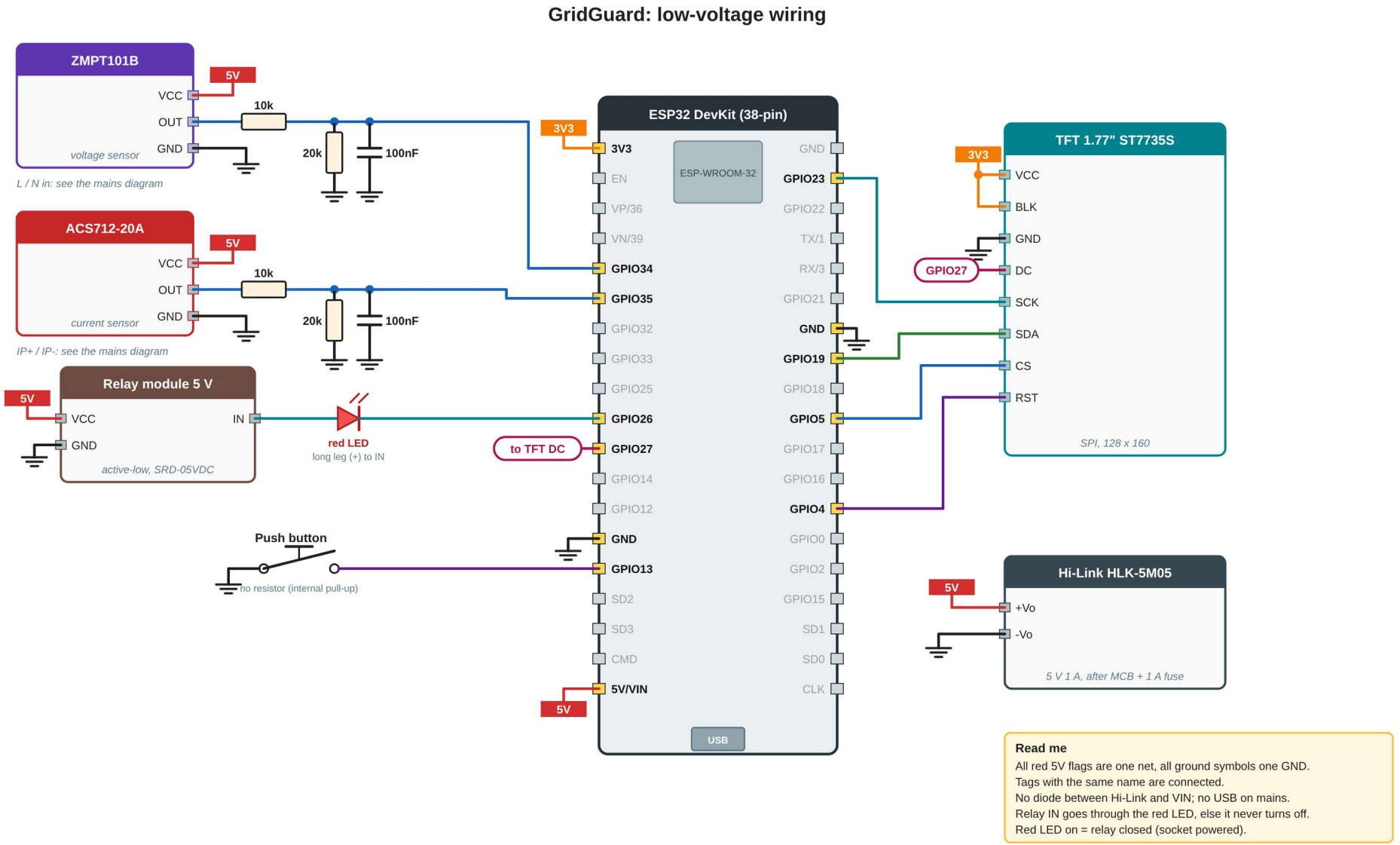
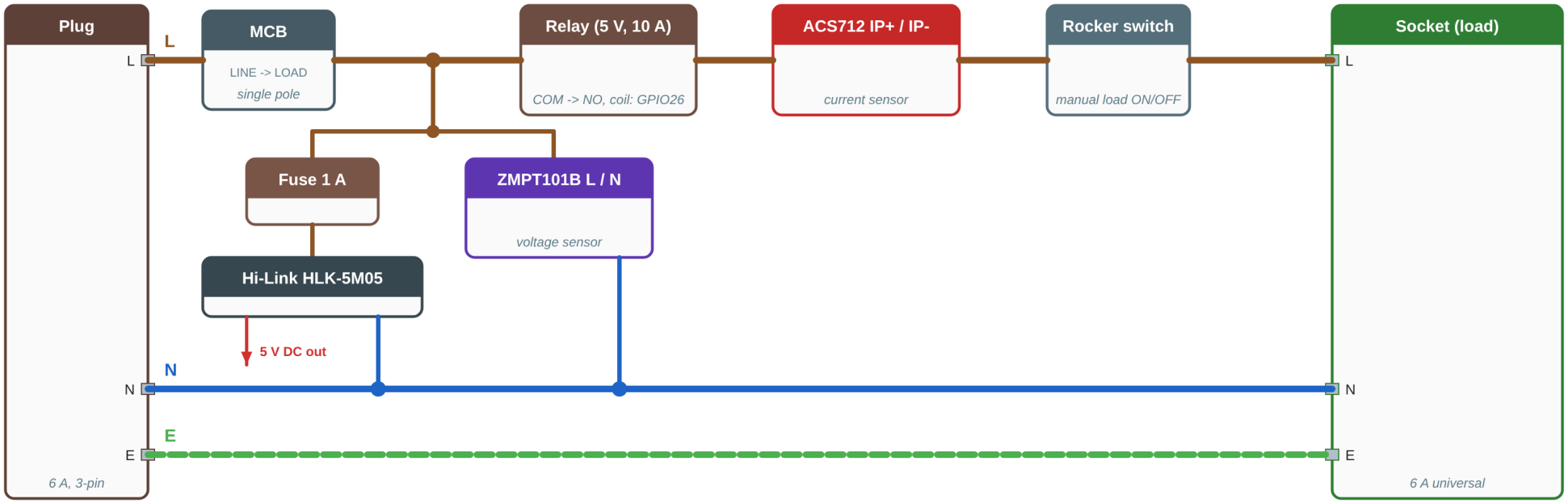


Figure 2. *GridGuard low-voltage wiring*

GridGuard: mains wiring (230 V)



Read me

Live only through the MCB, relay (COM -> NO), ACS712 and rocker switch. Neutral and earth go straight through. NO contact: no coil power = socket off.
Hi-Link and ZMPT101B are taken after the MCB and BEFORE the relay, so GridGuard keeps running and measuring after a trip.
1.5 sq mm wire for the load path, crimped lugs or ferrules, every module inside the closed box. Test the wall socket for swapped L and N first.

Figure 3. *GridGuard mains wiring (230 V)*

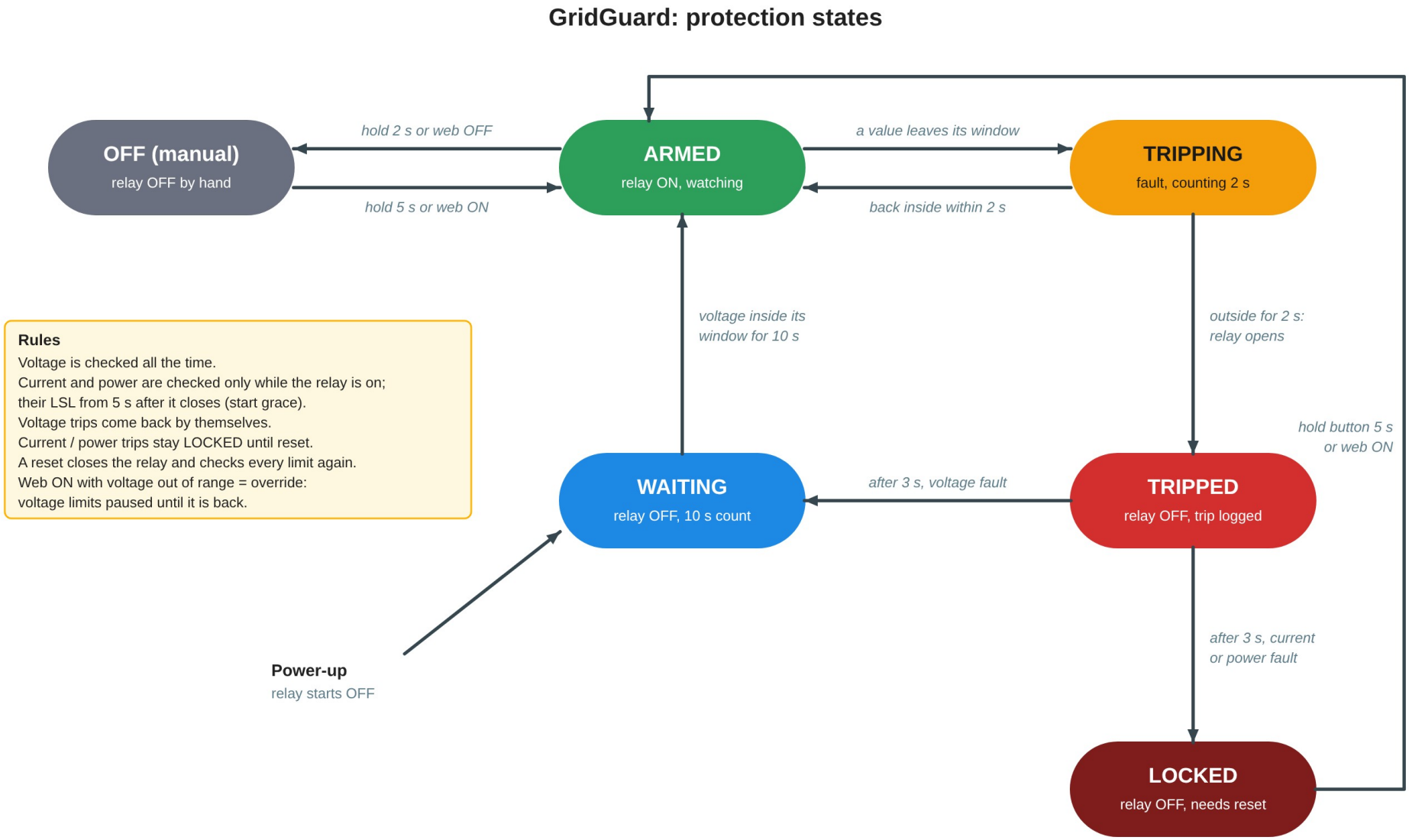


Figure 4. *GridGuard protection states*

6. Flashing and first setup

1. Install the Arduino IDE and the ESP32 board package (Espressif). Select board **ESP32 Dev Module**.
2. Set **Tools > Partition Scheme > Huge APP (3MB No OTA/1MB SPIFFS)**.
3. Install libraries: **Adafruit GFX Library** and **Adafruit ST7735 and ST7789 Library** (Library Manager); **ESPAsyncWebServer** and **AsyncTCP** by ESP32Async (download each as a ZIP, then Sketch > Include Library > Add .ZIP Library).
4. Unzip the provided **GridGuard.zip**. It contains the folder **GridGuard** with **GridGuard.ino**, **BaiJamjuree.h** and **web_index.h**; keep the three files together in that folder. Open GridGuard.ino.
5. Connect the ESP32 by USB (**mains unplugged**), choose the COM port and press Upload. Serial Monitor at 115200, line ending "Newline".
6. On a phone, join WiFi **GridGuard** / **gridguard** and open **http://192.168.4.1**.

```
Board              ESP32 Dev Module
Partition Scheme   Huge APP (3MB No OTA/1MB SPIFFS)
Libraries          Adafruit GFX, Adafruit ST7735 and ST7789, ESPAsyncWebServer, AsyncTCP
Files              GridGuard.zip -> GridGuard/GridGuard.ino + BaiJamjuree.h + web_index.h
Serial Monitor     115200 baud
```

7. Calibration and testing

Open the **Calibration** section of the web page (tap to expand). Values are saved in flash immediately. The same works from the Serial Monitor: `V 231.5`, `I 4.20`, `O -2.5`, `Z`, `F`, `D` (defaults), `?`.

7.1 Calibration steps

1. **Voltage:** measure the socket with a multimeter, type it under "Multimeter reads" and press **Match**. This sets the gain; the offset (−30 to +30 V) can also be typed.
2. **Current:** run a kettle or heater (at least 0.5 A) with the relay on, type the clamp meter reading under "Clamp meter reads" and press **Match**.
3. **Zero offset:** unplug the load and press **Measure zero** (about 4 s). It refuses if a load still draws current. An empty socket then reads 0.00 A.
4. **Direction:** press **Flip direction** if power reads wrong on a known load.
5. **Defaults** restores the factory calibration; the limits are not changed.

Table 6. Calibration values (defaults)

Value	Default	Range / meaning
Voltage gain	0.946	0.5 to 1.5; mains volts per mV at the pin
Voltage offset	0 V	−30 to +30 V added after the gain
Current trim	1.00	0.5 to 2.0; clamp meter A / shown A
Zero offset	0.195 A	0 to 1 A; idle noise of the current sensor
Direction	−1	+1 or −1

7.2 Protection tests

#	Check	Expected	Result
1	Power-up with good mains	STARTING 10 s, then ARMED, relay clicks on, red LED on	☐ pass ☐ fail
2	Voltage USL set just below the mains voltage	TRIP IN 2 s, TRIPPED, relay off, red LED off; WAITING; back on 10 s after USL is restored	☐ pass ☐ fail
3	Current USL 0.5 A with a kettle on	trips, then LOCKED (HOLD 5s RESET)	☐ pass ☐ fail
4	Hold the button 3 s, release	RESETTING countdown, then "Reset cancelled", stays LOCKED	☐ pass ☐ fail
5	Hold the button 5 s (USL back to 6 A)	relay on, ARMED	☐ pass ☐ fail
6	Current LSL 0.5 A, nothing plugged in	after 5 s grace and 2 s delay: under-current trip, LOCKED	☐ pass ☐ fail
7	Web OFF, then web ON	relay off (OFF manual), then on (ARMED)	☐ pass ☐ fail
8	Button hold 2 s with relay on	relay off (OFF manual)	☐ pass ☐ fail
9	ESP32 EN (reset) button	relay off while the ESP32 restarts	☐ pass ☐ fail
10	Power-cycle	limits and calibration kept; trip log cleared	☐ pass ☐ fail
11	Multimeter / clamp meter vs GridGuard	within 1 % / 2 % after calibration	☐ pass ☐ fail

8. Display and web page guide

8.1 TFT screens and button

A short press of the button shows the next screen; otherwise the screen stays where it is. A trip always brings Home back. Holding the button: with the relay on, 2 s turns it off; with the relay off, 5 s resets (RESETTING 5 4 3 2 1 on the screen).

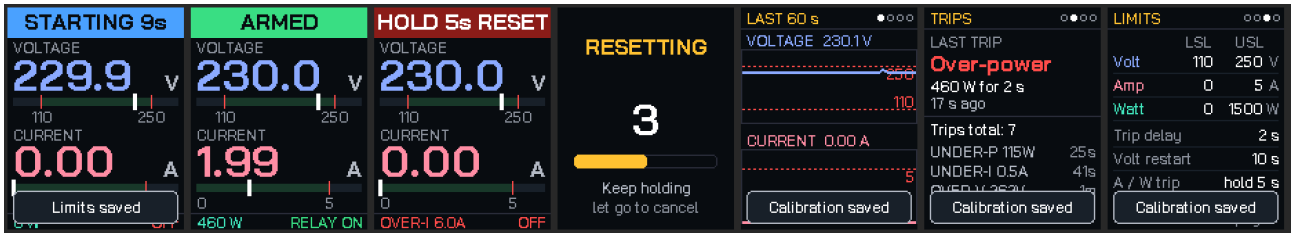
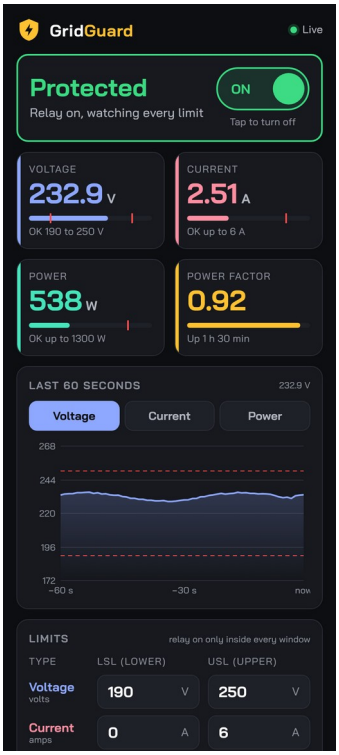


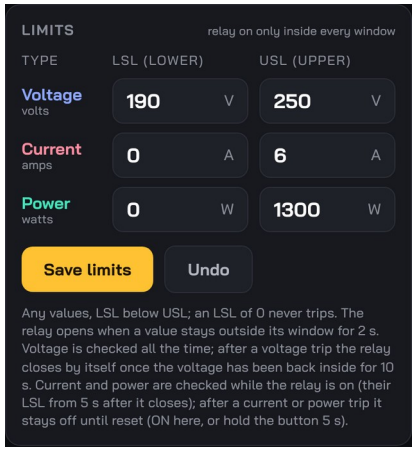
Figure 5. *TFT screens: STARTING, ARMED, LOCKED, RESETTING countdown, Graph, Trips, Limits*

Screen	Shows
Home	status bar (ARMED green, TRIP IN amber, TRIPPED red, WAITING / STARTING blue, LOCKED dark red, OFF grey), voltage and current with bars showing the LSL–USL window, power and relay state
Graph	last 60 s of voltage and current with the limits as dashed lines
Trips	last trip with reason, value and time, total since power-up, earlier trips
Limits	LSL and USL of voltage, current and power; trip and restart delays
Connect	WiFi name, password, address, phones connected, firmware version, button help

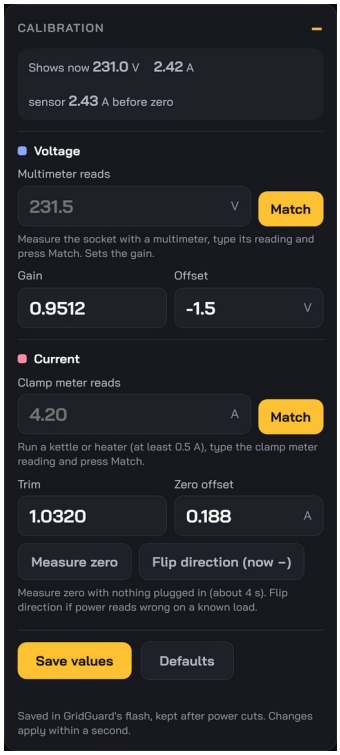
8.2 Web dashboard



Status, relay switch, live values



Limits table (LSL / USL)



Calibration

Figure 6. *GridGuard web page*

Part	What it does
Status card	state in words with the reason and countdown; the ON/OFF switch turns the relay on or off (OFF asks to confirm)
Live tiles	voltage, current, power with their windows marked, power factor and uptime
Last 60 seconds	graph of voltage, current or power with the limits
Limits	LSL and USL for each quantity; Save limits / Undo
Calibration	match to meters, gain, offset, trim, zero offset, measure zero, flip, defaults
Trip log	last 10 trips with reason, value and time ago

Web ON while the voltage is out of range

Tapping ON on the web page while the voltage is outside its window starts an **override**: the relay closes and the voltage limits are ignored until the voltage comes back inside them. Current and power protection stay on. The button reset never overrides: it checks all limits straight away.

9. Troubleshooting

Problem	Likely cause	Fix
TFT stays white	SDA / SCK swapped or on other pins than the sketch; RST / DC / CS loose	check the wiring against Table 4; BLK to 3V3
Picture shifted, mirrored or red / blue swapped	different ST7735 panel variant	try INITR_GREENTAB or INITR_REDTAB in initR()
Relay always ON, even when off	active-low module: a released 3.3 V pin still looks low	put a red LED in series with IN (long leg to IN); RELAY_ACTIVE_LOW = true
Relay ticks or buzzes	IN driven to 3.3 V, half on	same fix: series LED and the current firmware
Relay LED lights but no click	12 V relay (SRD-12VDC) on 5 V, or JD-VCC not powered	use a 5 V relay; on JD-VCC boards power JD-VCC from 5 V or refit the jumper
Works on USB, resets on the Hi-Link	series diode dropped VIN to about 3.6 V	no diode between the Hi-Link and VIN
Compile error "declared void"	an older copy of the sketch	use GridGuard.ino from the provided zip
"Sketch too big"	default partition	Huge APP (3MB No OTA/1MB SPIFFS)
Relay never closes after power-up (BAD SUPPLY)	voltage outside its LSL–USL window, or voltage not calibrated	check the limits and calibrate the voltage
Keeps showing HOLD 5s RESET	a current or power trip; locked on purpose	fix the cause, then hold 5 s or tap ON
Under-current trips right after reset	current LSL above the load's real current	lower or zero the current LSL; the start grace is 5 s
Current match refused	less than 0.5 A flowing	use a kettle or heater with the relay on
Measure zero refused	a load is still connected	unplug the load and try again
Small loads read 0 A	below the noise floor (about 0.2 A)	expected with the 20 A sensor